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YINHAN HE, University of Virginia, Charlottesville, VA, United States

CHEN CHEN, University of Virginia, Charlottesville, VA, United States

SONG WANG, University of Virginia, Charlottesville, VA, United States

GUANGHUI MIN, University of Virginia, Charlottesville, VA, United States

JUNDONG LI, University of Virginia, Charlottesville, VA, United States

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Demystify Epidemic Containment in Directed Networks: Theory and Algorithms

Yinhan He
University of Virginia
Charlottesville, VA, USA
nee7ne@virginia.edu

Chen Chen
University of Virginia
Charlottesville, VA, USA
zrh6du@virginia.edu

Song Wang
University of Virginia
Charlottesville, VA, USA
sw3wv@virginia.edu

Guanghui Min
University of Virginia
Charlottesville, VA, USA
jjm8vr@virginia.edu

Jundong Li
University of Virginia
Charlottesville, VA, USA
jl6qk@virginia.edu

Abstract

Epidemic containment has long been a crucial task in many high-stake application domains, ranging from public health to misinformation dissemination. Existing studies for epidemic containment are primarily focused on undirected networks, assuming that the infection rate is constant throughout the contact network regardless of the strength and direction of contact. However, such an assumption can be unrealistic given the asymmetric nature of the real-world infection process. To tackle the epidemic containment problem in *directed networks*, simply grafting the methods designed for undirected network can be problematic, as most of the existing methods rely on the orthogonality and Lipschitz continuity in the eigensystem of the underlying contact network, which do not hold for directed networks. In this work, we derive a theoretical analysis on the general epidemic threshold condition for directed networks and show that such threshold condition can be used as an optimization objective to control the spread of the disease. Based on the epidemic threshold, we propose an asymptotically greedy algorithm **DINO** (**DI**rected **N**etw**O**rk epidemic containment) to identify the most critical nodes for epidemic containment. The proposed algorithm is evaluated on real-world directed networks, and the results validate its effectiveness and efficiency. The code is available at https://github.com/YinhanHe123/DINO/.

CCS Concepts

• **Networks** → *Network dynamics*; **Network algorithms**; • **Computing methodologies** → Modeling methodologies; • **Theory of computation** → **Graph algorithms analysis**.

Keywords

Directed Networks, Network Epidemiology, Epidemic Containment

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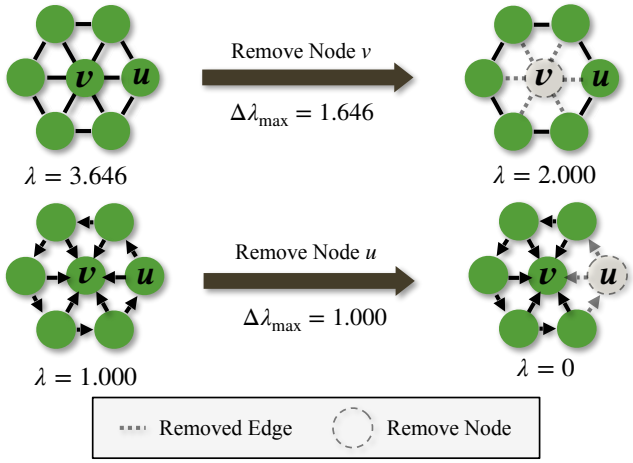


Figure 1: An example of node selection to achieve the largest spectral radius decrease for undirected (left) and directed networks (right).

1 Introduction

Epidemic containment [12, 32, 42], which aims to find the best strategies to confine the spread of diseases in the contact network, has become an increasingly important task in public health. Specifically, policymakers typically want to vaccinate individuals who may potentially become superspreaders in the network, as immunization of these individuals can restrict the transmission of the disease to the maximum extent, therefore reducing the overall impact of the epidemic. Epidemic containment strategies can be applied naturally in other application domains, including containing the spread of cyber threats in cybersecurity [41] or the dissemination of misinformation [15]. Over the years, extensive research efforts have been dedicated to understanding the fundamental principles of epidemic spread process in undirected networks [10, 11, 14, 20, 23, 38, 44, 46, 47]. Specifically, many studies have identified a crucial number named *epidemic threshold* in the epidemic control process. When the *spectral radius of the network* is above the epidemic threshold, the disease spread can no

longer be contained and will start spreading rapidly within a population [38, 44, 47]. Here, *spectral radius* of a network is the largest eigenvalue magnitude of its adjacency matrix. Drawing on this crucial finding, an effective way to contain an epidemic is to immunize (i.e., remove) the nodes whose removal will have the greatest impact in decreasing the spectral radius of the network.

Although effective, the undirected contact network assumption in previous research efforts can be unrealistic in real applications. This assumption implies that, within a connected pair of nodes, each node has an equal probability of being infected by its counterpart, regardless of the health condition of each individuals. However, a current study shows that, for most epidemics, it is possible that one can easily infect an immunocompromised individual, while the reverse might be significantly less probable [39]. In such assymetric networks, directly applying existing epidemic containment strategies designed for undirected networks may lead to suboptimal solutions as illustrated in Fig. 1. In particular, suppose that we are restricted to vaccinating (i.e., removing) only one node to control the disease spread process. In the directed network, removing the node u or any peripheral nodes can effectively cut the transmission path of the disease. On the other hand, if we ignore the edges' directions and remove the node that leads to largest spectral radius reduction, node v would be selection in this case. Mapping back to the directed network, the removal of sink node v however would not affect the disease spread process at all.

Despite the importance of considering edge directions for epidemic containment, finding the best containment strategy on directed networks is non-trivial and challenging. Specifically, the challenges are two-fold: (1) ***Gap towards Epidemic Threshold:*** The relationship between the spectral radius and epidemic threshold has been well-established for undirected networks [40]. While an initial study has explored this relationship in the context of directed graphs, it was limited to a specific epidemic model [28]. Unlike undirected networks, the adjacency matrices of directed networks can have complex eigenvalues. This mathematical complexity introduces significant challenges in determining the epidemic threshold for directed networks across various epidemic models. (2) ***Gap towards Epidemic Containment Methods:*** Existing epidemic containment methods are predominantly designed for undirected networks by leveraging matrix perturbation theory [10, 11, 30, 46]. Such methods can not be extended to directed networks as suggested in [31], due to the lack of the two properties: (i) *Orthogonal Basis:* The eigenvectors of the network adjacency matrix used to calculate the spectral radius do not form an orthonormal basis on directed networks. This prevents the applications of previous works, such as the Davis-Kahan theorem [48], which relies on eigenvector orthogonality to provide bounds on eigenspace perturbations. (ii) *Lipschitz Continuity:* The eigenvalue perturbation of the network adjacency matrix is not Lipschitz continuous on directed networks [27]. This means that a tiny perturbation of the matrix could make a large change in the spectrum, making the spectral radius approximation no longer precise.

In this work, we take a first step towards the epidemic containment problem on directed networks:

- ***Theoretical Analysis:*** We present three crucial results: (1) *Spectrum of Subgraphs:* We demonstrate that by decreasing the spectral radius of a particular subgraph of a directed network, the spectral radius of the entire network is reduced. (2) *Spectral Radius Approximation:* Under the assumption that real-world directed networks can be modeled by the Directed Chung-Lu Random Network model and an assumption on the network connectivity, we provide a tight bound for the spectral radius of directed networks. (3) *Epidemic Threshold:* We generalize the epidemic threshold from undirected networks to directed networks. We also prove that the epidemic will die out by itself if the network spectral radius is smaller than the epidemic threshold.
- ***Algorithms:*** Based on the above theory, we present an *asymptotically greedy algorithm* called **DINO** (**DI**rected **N**etw**O**rk epidemic containment) to find k nodes whose removal can maximally contain the epidemic on directed networks. In particular, DINO does not rely on matrix perturbation, thus solves the aforementioned challenges.
- ***Evaluations:*** We conduct comprehensive experiments on multiple real-world directed networks to verify the effectiveness and efficiency of the proposed DINO in decreasing the network spectral radius, thus controlling epidemic propagation.

2 Preliminaries

Epidemic Spread Models and Networks. Epidemic spread models are non-linear dynamic systems that simulate the spread of epidemics [40]. Individuals can be classified into susceptible (S), infected (I), and recovered or vigilant (R or V) groups. Susceptible individuals are healthy but at risk of infection. Infected individuals are sick and can transmit the disease. Recovered or vigilant individuals are healthy individuals no longer susceptible to the disease. Various models, such as *SIS*, *SIR*, *SEIR*, and *SIRS* [1, 25], have been proposed to simulate different epidemic spread patterns. Recent studies have shown that the directed Chung-Lu random network model serves as an excellent simulation for real-world epidemic spread networks [3, 17, 19]. A directed Chung-Lu network is constructed based on a network distribution $G(\mathbf{d}^{in}, \mathbf{d}^{out})$, where $\mathbf{d}^{in} \in \mathbb{N}^n$ and $\mathbf{d}^{out} \in \mathbb{N}^n$ indicate the expected in-degree and out-degree sequences of the network, respectively. When generating this network, every potential directed edge (i, j) appears independently with probability $\frac{d_i^{out} d_j^{in}}{vol(\mathcal{G})}$. Here, d_i^{out} and d_j^{in} are the ith and jth entry of $\mathbf{d}^{out}$ and $\mathbf{d}^{in}$, respectively. The network's volume $vol(\mathcal{G})$ is defined as the sum of its in-degree and out-degree sequences $vol(\mathcal{G}) = \sum_i d_i^{in} = \sum_i d_i^{out}$. Studying the properties of epidemic spread on real-world networks can be effectively achieved by examining those properties on the directed Chung-Lu network [13].

Epidemic Threshold on Undirected Networks. While different epidemic spread models exist, many of them can be unified under a general framework proposed in [40]. This framework, called $S^* I^2 V^*$, consists of a sequence of susceptible states denoted by S^*, two infected states denoted by I^2, and a sequence of vigilant states denoted by V^*. We denote each susceptible state in S^* as S_y ($y \in \{1, ..., |S^*|\}$), and each vigilant state in V^* as V_x ($x \in \{1, ..., |V^*|\}$). When an epidemic of the form $S^* I^2 V^*$ spreads in a network, the infected nodes have a certain probability of infecting their neighbors at each time step. Meanwhile, susceptible and vigilant nodes have certain probabilities of transitioning to other states in the S^* and V^* sequences,

respectively. We define epidemic model parameters of $S^*I^2V^*$ following [40]. Specifically, I_0 is the proportion of people who are initially infected in the population, $\alpha_{X,Y}$ is the probability of transmission from state X to Y; β_1 and β_2 are the transmission probabilities for the first infected state E and the second state I in I^2, respectively; $p^*_{S_y}$ is the probability of the network node being present in S_y at the equilibrium point of the system. Given these, an important number called *epidemic threshold*, is defined. It is a combination of the epidemic model's parameters. Specifically, the epidemic threshold can be expressed as $\frac{(1-\alpha_{II})(1-\alpha_{EE})-\alpha_{IE}\alpha_{EI}}{p^*_S(\beta_1(1-\alpha_{II})+\beta_2\alpha_{EI})}$, where $p^*_S = \Sigma^n_y p^*_{S_y}$ is the summation of probabilities that one node presents in any susceptible state. The epidemic will finally be eliminated if network *spectral radius* is smaller than the threshold.

Based on the preliminaries above, we can formally define the concept of epidemic threshold on directed networks and the problem of epidemic containment on directed networks below.

Definition 2.1. (**Epidemic Threshold on Directed Networks**) Given a directed network $\mathcal{G}$ and a general epidemic propagation model $S^*I^2V^*$, the epidemic threshold is a number composed of epidemic spread model parameters such that the epidemic will die out if and only if the network spectral radius is smaller than it.

Problem 1. ***(Epidemic Containment Problem)*** *Given: a directed network $\mathcal{G}$ with n nodes and a positive integer k where $k < n$. Find: a subset of k nodes such that removing those nodes minimizes the spectral radius of the network adjacency matrix.*

3 Theory

3.1 Spectral Radius of Subgraphs

Here, we present an important theorem uncovering the fundamental spectrum relationship between a directed network and its strongly connected components.

Theorem 3.1. *Let $\mathcal{G}$ be a directed network with its strongly connected components $\{\mathcal{S}_1, \ldots, \mathcal{S}_t\}$. Define the key strongly connected component (KSCC) $\mathcal{S}^* = \arg\max_{\mathcal{S}\in\{\mathcal{S}_1,\ldots,\mathcal{S}_t\}} \rho_{\mathcal{S}}$, where $\rho_{\mathcal{S}}$ is the spectral radius of $\mathcal{S}$, then the spectral radius $\rho_{\mathcal{G}}$ of $\mathcal{G}$ is $\rho_{\mathcal{G}} = \rho_{\mathcal{S}^*}$.*

Proof. We denote two strongly connected components $\mathcal{S}_i$ and $\mathcal{S}_j$ to have the relationship $\mathcal{S}_i \to \mathcal{S}_j$ if an edge connects $\mathcal{S}_i$ to $\mathcal{S}_j$. If $\mathcal{S}_i \to \mathcal{S}_j$, then $\mathcal{S}_j \nrightarrow \mathcal{S}_i$, otherwise $\mathcal{S}_i$ and $\mathcal{S}_j$ are part of a larger strongly connected component. With this observation, we can sort $\{\mathcal{S}_1, ..., \mathcal{S}_t\}$ such that for any $\mathcal{S}_i$ and $\mathcal{S}_j$, if $\mathcal{S}_i \to \mathcal{S}_j$, then $i < j$. Therefore, the adjacency matrix can be written as

$$\begin{pmatrix} A_{\mathcal{S}_1} & A_{\mathcal{S}_1\to\mathcal{S}_2} & A_{\mathcal{S}_1\to\mathcal{S}_2} & \cdots & A_{\mathcal{S}_1\to\mathcal{S}_t} \\ 0 & A_{\mathcal{S}_2} & A_{\mathcal{S}_2\to\mathcal{S}_3} & \cdots & A_{\mathcal{S}_2\to\mathcal{S}_t} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \cdots & 0 & A_{\mathcal{S}_t} \end{pmatrix}.$$

We can calculate the determinant of this matrix using the block matrix determinant formula:

$$\det(A_{\mathcal{G}} - \lambda I) = \det(A_{\mathcal{S}_1} - \lambda I)...(A_{\mathcal{S}_t} - \lambda I) \tag{1}$$

Note that for any matrix $M \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, ...\lambda_k \in \mathbb{C}$ and corresponding multiplicity $r_1, ..., r_k$, characteristic polynomial could be written as $\Pi^k_{i=1}(x - \lambda_i)^{r_i}$. This shows that the roots of the characteristic polynomial of a matrix are composed of all its eigenvalues. Consequently, graph $\mathcal{G}$'s spectral radius is the same as that of its key strongly connected component, i.e., the strongly connected component with the largest magnitude of eigenvalues. □

Theorem 3.1 implies that to decrease the spectral radius of one directed network, we just need to decrease that of its key strongly connected component.

3.2 Spectral Radius Approximation

Under the assumption that a real-world directed epidemic spread network can be modeled by the Directed Chung-Lu Random Network model [13], we provide a tight bound for the network spectral radius conditioning on the network being strongly connected:

Theorem 3.2. *For a directed Chung-Lu network $\mathcal{G}$, if (1) $\frac{\mathbf{d}^{in}\cdot\mathbf{d}^{out}}{vol(\mathcal{G})} > 1$, (2) define $p_{max} := \frac{\mathbf{d}^{in}_{max}\mathbf{d}^{out}_{max}}{vol(\mathcal{G})}$ and it satisfies $p_{max} = O(N^{-1})$, define $p_{min} := \frac{\mathbf{d}^{in}_{min}\mathbf{d}^{out}_{min}}{vol(\mathcal{G})}$, where $\mathbf{d}^{in}_{max}$ and $\mathbf{d}^{out}_{max}$ represents the largest entry of the vector $\mathbf{d}^{in}$ and $\mathbf{d}^{out}$ respectively and $\mathbf{d}^{in}_{min}$ and $\mathbf{d}^{out}_{min}$ represents the smallest entry of the vector $\mathbf{d}^{in}$ and $\mathbf{d}^{out}$ respectively, (3) $\mathcal{G}$ is strongly connnected, then for any constant $C > 1$, there exist $N' \in \mathbb{N}$ and $p^* \in (0, p_{max})$ such that when $N > N'$ and $p_{min} \in (p^*, p_{max})$ and $\epsilon = O\left(\frac{(\log N)^2}{N^{1/2}}\right)$, we have*

$$(1 - \frac{1}{\sqrt{C}}\epsilon)\frac{\mathbf{d}^{in}\cdot\mathbf{d}^{out}}{vol(\mathcal{G})} \le \rho_G \le (1+\epsilon)\frac{\mathbf{d}^{in}\cdot\mathbf{d}^{out}}{vol(\mathcal{G})}. \tag{2}$$

Proof. We delineate our proof strategy as follows. Based on Lemma 1 in [8], for every positive integer r, we have the following bounds on the maximum eigenvalue of A:

$$\frac{trace(A^r)}{N} \le \rho(A^r) = \rho(A)^r \le \mathbf{1}^T A^r \mathbf{1} \tag{3}$$

where $\mathbf{1}$ is the vector of ones. It is easy to show that $trace(A^r) = \sum_{j=1}^N e_j^T A^r e_j$ (where e_j is the standard unit vector) represents the number of cycles of length r, and $\mathbf{1}^T A^r \mathbf{1}$ represents the number of paths of length r. Therefore, to control the spectral radius of the graph adjacency matrix, it suffices to estimate the number of cycles/paths within the graph.

We express the lower bound of Equ. 3 by Bayes theorem:

$$\begin{aligned} &\mathbb{E}(\frac{trace(A^r_{\mathcal{G}})}{N}|\mathcal{G}\in\text{SCG}) \\ =&\mathbb{E}(C_r|\mathcal{G}\in\text{SCG}) \\ =&\Sigma_{\Omega_r}P(\Omega_r \text{ exists}|\mathcal{G}\in\text{SCG}) \\ =&\Sigma_{\Omega_r}\frac{P(\Omega_r \text{ exists})\times P(\mathcal{G}\in\text{SCG}|\Omega_r \text{ exists})}{P(\mathcal{G}\in\text{SCG})}. \end{aligned} \tag{4}$$

In this context, $C_r := \frac{trace(A^r_{\mathcal{G}})}{N}$ represents the number of cycles of length r, and Ω_r denotes a specific cycle of length r. $\mathbb{E}(C_r|\mathcal{G}\in\text{SCG})$ denotes the expected number of cycles of length r, given that $\mathcal{G}$ is strongly connected. $P(\Omega_r \text{ exists}|\mathcal{G}\in\text{SCG})$ represents the probability that the specific cycle Ω_r exists, given that $\mathcal{G}$ is strongly connected. $P(\mathcal{G}\in\text{SCG}|\Omega_r \text{ exists})$ represents the probability that $\mathcal{G}$ is strongly connected, given that Ω_r exists.

Now, our aim is to examine how the condition that $\mathcal{G}$ is strongly connected can provide information to measure the cycle numbers in the graph $\mathcal{G}$ based on Equ. 4. To perform the examination, we initially prove that $\frac{P(\mathcal{G}\in\text{SCG}|\Omega_r \text{ exists})}{P(\mathcal{G}\in\text{SCG})} > 1$ holds in Equation 4. We prove it because this result directly implies that the strong connectivity of $\mathcal{G}$ leads to a tighter lower bound on $\rho(A)$ in Equ. 3. More quantitatively, we can show this lemma:

Lemma 3.3. *Denote $\mathcal{G}$ as a realization of a strongly connected random Chung-Lu graph with expected degree sequence $\mathbf{d} = (\mathbf{d}^{in}, \mathbf{d}^{out}) \in \mathbb{Z}^{N\times 2}$. Let $p_{max} = \frac{\mathbf{d}^{in}_{max}\cdot\mathbf{d}^{out}_{max}}{vol(\mathcal{G})}$, $p_{min} = \frac{\mathbf{d}^{in}_{min}\cdot\mathbf{d}^{out}_{min}}{vol(\mathcal{G})}$ and $vol(\mathcal{G}) = \sum_i \mathbf{d}^{in}_i = \sum_i \mathbf{d}^{out}_i$. When $p_{max} = O(\frac{1}{N})$ and $p_{max} < \frac{1}{N}$, for any $C > 1$, there exists $N' \in \mathbb{Z}$ and $p^* \in (0, p_{max})$; when $N > N'$ and $p_{min} \in (p^*, p_{max})$, $\mathbb{E}(C_r|\mathcal{G} \in \textbf{SCG}) > C \cdot \mathbb{E}(C_r)$, where C_r is the number of cycles of length r.*

This lemma shows that when the number of vertices N in $\mathcal{G}$ is sufficiently large and p_{min} is close enough to p_{max}, the expected number of cycles of length r in the graph is significantly higher under the condition of strong connectivity compared to that of general directed graphs. By replacing $\mathbb{E}(C_r)$ with $\mathbb{E}(C_r|\mathcal{G} \in \textbf{SCG})$ in Equ. 43 of Theorem 5 in [8], we obtain the tight lower bound presented in our theorem. The upper bound follows [8]. □

This theorem indicates that the spectral radius approximation converges fast for strongly connected directed networks, resulting in precise spectral radius estimation for strongly connected networks. The tight bound given by Theorem 3.2 sets a solid theoretical foundation for our epidemic containment algorithm introduced in next section, since it relies on estimating the spectral radius of strongly connected components and further selecting immunizing nodes from these components.

3.3 Epidemic Threshold on Directed Networks

Several works have explored the epidemic threshold for directed networks [28, 34]. [28] focuses on the specific epidemic model *SIS*, while [34] is only applicable to small-size graphs. In our work, we generalize the definition of the epidemic threshold from undirected networks to directed networks of any size. In other words, we consider the general epidemic spread model $S^*I^2V^*$[40] on a weighted directed network, which can represent a broad spectrum of various epidemic models, effectively improving the applicability of the epidemic threshold in different epidemic scenarios.

Theorem 3.4. ***(Epidemic Threshold on Directed Network)*** *For a $S^*I^2V^*$ epidemic on a weighted directed network $\mathcal{G}$ (with adjacency matrix $A_{\mathcal{G}}$ and spectral radius $\rho_{\mathcal{G}}$), the epidemic spread dies out if $\rho_{\mathcal{G}} < C$. The largest such C is defined as the epidemic threshold, which is a constant depending on the epidemic model.*

Proof. We model the spread of disease in the network as a nonlinear dynamic system g and prove its stability by evaluating the eigenvalue of its Jacobian matrix. The proof generally follows the structure in [40] but tailored for *directed networks*. Specifically, we write the system as $\mathbf{P}_{t+1} = g(\mathbf{P}_t)$, where $\mathbf{P}_t$ describes each node's state in the system at time step t. According to the asymptotic stability condition of a dynamic system, the epidemic will asymptotically contained if norm of eigenvalues of $\nabla g(x)$ is less than 1. This condition achieves if the solutions' norm (i.e., $||\lambda_J||$) of the following Equ. 5 is smaller than 1:

$$\lambda_J^2 - \lambda_J(\alpha_{EE} + \alpha_{II} + p_s^*\beta_1\rho_{\mathcal{G}}) + (\alpha_{II}\alpha_{EE} - \alpha_{IE}\alpha_{EI} + p_s^*\rho_{\mathcal{G}}(\beta_1\alpha_{II} - \beta_2\alpha_{EI})) = 0. \tag{5}$$

Here, α and β with any subscripts are real-number constant parameters of the epidemic model. As $\rho_{\mathcal{G}}$ is essentially an eigenvalue of a directed graph's adjacency matrix, it may be a complex number, which significantly complicate the case to solve the asymptotic stability condition than that is in [40]. However, as A is a non-negative matrix, i.e., each entry of the matrix is larger or equal to 0, then by Frobenius-Perron theorem, we know that $\lambda_A \in \mathbb{R}^+$ is a real number. Therefore, we may directly utilize the property of real-coefficient quadratic equation and go through the same derivation of (27)-(29) in [40], the spectral radius of the network $\mathcal{G}$ is required to satisfy the following condition to achieve the system asymptotic stability (i.e., the epidemic containment):

$$\rho_G < p_S^*\left(\frac{\beta_1(1-\alpha_{II}) + \beta_2\alpha_{EI}}{(1-\alpha_{II})(1-\alpha_{EE}) - \alpha_{IE}\alpha_{IE}}\right). \tag{6}$$

Additionally, it is worth noting that the conclusion naturally applys to weighted directed networks since we only require the adjacency matrix $A_{\mathcal{G}}$ of network $\mathcal{G}$ is non-negative. □

4 Algorithm and Analysis

According to Theorem 3.4, an epidemic spreading on a directed network will be eliminated if the network spectral radius is smaller than the epidemic threshold. Therefore, in this section, we introduce an effective and efficient algorithm, named DINO, for epidemic containment on directed networks by immunizing (i.e., removing) nodes that cause maximum network spectral radius reduction.

4.1 DINO: The Proposed Algorithm

We introduce the DINO Algorithm, which is described in Algorithm 1. DINO aims to pick a set of nodes with a limited budget of k, whose removal from the directed network will significantly contain the spread of epidemics. Specifically, DINO operates sequentially by selecting one node from the key strongly connected component (KSCC) with the highest score (node score function $F(\cdot)$ is defined later). The selected node will then be removed from the network, and the process repeats k rounds until the budget is used up.

Selecting Node within KSCC. Based on Theorem 3.1, only by removing nodes from the key strongly connected component (KSCC) of a directed network can achieve the goal of decreasing the spectral radius of the original network. Therefore, we design a sequential framework to ensure that we remove nodes from the KSCC in each round. Here, we give a detailed description of the proposed framework. We begin by identifying the non-trivial strongly connected components (i.e., strongly connected components with node number larger than two) on the epidemic spread network. These strongly connected components are then arranged in descending order based on their spectral radius; the resulting strongly connected components sequence is represented as $S = \{\mathcal{S}_1, \mathcal{S}_2, \ldots, \mathcal{S}_t\}$. From the sequence S, we extract the first component $\mathcal{S}_1$ and subsequently remove it from S. Within $\mathcal{S}_1$, we identify and remove the node with the highest score, denoted by v. After the removal of node v and its associated edges on the network, we find the strongly connected

Algorithm 1 Directed Network Epidemic Containment Algorithm (DINO)

Input: (1) a directed network $\mathcal{G}$; (2) our node score function $F(\cdot)$; and (3) node immunization budget k
Output: the index list L of size k.
initialize L to be empty;
compute $S \leftarrow$ Find non-trivial strongly connected components $\{\mathcal{S}_1, ..., \mathcal{S}_t\}$, sort reversely by spectral radius;
for $l = 1, 2, \ldots, k$ **do**
  while $S \neq \emptyset$ **do**
    $\mathcal{S}_1 \leftarrow$ the first element in S; $S \leftarrow S \setminus \mathcal{S}_1$; $v \leftarrow \arg\min_{u \in \mathcal{G}} F(u)$, $L \leftarrow L \cup \{v\}$;
    compute $\{\tilde{\mathcal{G}}_1, ..., \tilde{\mathcal{G}}_{\tilde{t}}\} \leftarrow$ nontrivial strongly connected components set of $\mathcal{S}_1 \setminus v$;
    $S \leftarrow$ insert $\{\tilde{\mathcal{G}}_1, ..., \tilde{\mathcal{G}}_{\tilde{t}}\}$ by spectral radius.
  end while
end for
Return L

components of $\mathcal{S}_1 \setminus \{v\}$ (multiple new SCCs could appear because of the removal of node v and its surrounding edges). These new SCCs are then inserted back to S, positioned according to their their network spectral radius.

Remarks. Constraining the node selection scope within the KSCC in each round has two benefits: (1) *Effectiveness*: By Theorem 3.1, only by removing a node from the KSCC of a network can we decrease the spectral radius of the whole directed network, thus enhancing the effect of epidemic containment. (2) *Efficiency*: As reported by [35], the majority of real-world social networks, including those used to study epidemic spread, have a key strongly connected component that is approximately 40% of the total size. By limiting node immunization to KSCC, we dramatically reduce the search space for nodes to immunize, thus greatly improving the efficiency of the optimization algorithm. **Node Score Function.** Based on the epidemic threshold of directed networks, we select the node to be immunized as the one whose removal causes the largest decrease in the graph spectral radius. However, directly calculating the spectral radius decrease for each node is computationally expensive ($O(N^3)$). Therefore, we use an efficiently calculated node score function to approximate the node whose removal introduces the largest spectral radius decrease. Bounding the spectral radius for general directed graphs is challenging, so we assume the epidemic network is an implementation of directed Chung-Lu networks. According to Theorem 3.2, we can estimate the spectral radius decrease caused by removing node v with $\frac{\mathbf{d}^{in}_{\mathcal{G}} \cdot \mathbf{d}^{out}_{\mathcal{G}}}{vol(\mathcal{G})} - \frac{\mathbf{d}^{in}_{\mathcal{G}\setminus\{v\}} \cdot \mathbf{d}^{out}_{\mathcal{G}\setminus\{v\}}}{vol(\mathcal{G}\setminus\{v\})}$, where $\mathcal{G} \setminus \{v\}$ represents the graph induced by $\mathcal{G}$ after removing node v. Since $\frac{\mathbf{d}^{in}_{\mathcal{G}} \cdot \mathbf{d}^{out}_{\mathcal{G}}}{vol(\mathcal{G})}$ is constant for various nodes, we design the node score function as $F(v) := \frac{\mathbf{d}^{in}_{\mathcal{G}\setminus\{v\}} \cdot \mathbf{d}^{out}_{\mathcal{G}\setminus\{v\}}}{vol(\mathcal{G}\setminus\{v\})}$. Selecting the node that decreases the spectral radius the most is equivalent to selecting the node that minimizes the score function, i.e., $\arg\min_{v \in \mathcal{G}} F(v)$. Although directly computing $F(v)$ costs time complexity of $O(N^2)$, the computation cost can be reduced to $O(N)$ by computing $\mathbf{d}^{in}_{\mathcal{G}}$, $\mathbf{d}^{out}_{\mathcal{G}}$, and $S_{\mathcal{G}}$ once and share throughout the computation of $F(v)$ for all nodes.

Remarks. Leveraging the designed score function has two benefits: (1) *Effectiveness*: Our node score function asymptotically approximates the reduced spectral radius resulting from the removal of a node from the original network. (2) *Efficiency*: Traditionally, explicitly calculating the spectral radius reduction such as employing the Arnoldi Algorithm [2] for every node will take at least $O(N^2)$ time. In contrast, our proposed node score function depends solely on the degrees of the nodes, allowing us to compute the node score for all nodes in linear time.

4.2 Effectiveness and Efficiency

The two theorems below prove that the Algorithm 1 asymptotically approximates greedy optimal with a approximation performance bound with the global optimal algorithm. In addition, we prove that our DINO algorithm has linear-time complexity.

Theorem 4.1. ***(Effectiveness)****We assume a real-world epidemic spread network can be modeled as a directed Chung-Lu network $\mathcal{G}$. We assume that all subgraphs of $\mathcal{G}$ with nodes more than $N - k$ (N large enough and $K \ll N$) satisfies conditions of Theorem 3.2. Given a node immunization budget k and our designed node score function $F(\cdot)$, let $\mathcal{L}_1$ denote the set of nodes immunized (that is, removed) by Algorithm 1, and $\mathcal{L}_2$ denote the set of nodes removed by the greedy algorithm (that is, sequentially removing the node with the largest reduction in spectral radius). Then we have $\lim_{n\to\infty} \rho(\mathcal{G} \setminus \mathcal{L}_1) = \rho(\mathcal{G} \setminus \mathcal{L}_2)$. Further, there exists a approximation performance bound between our DINO and the global optimal: $\Delta\rho_{DINO} \geq \frac{1}{\alpha}[1 - (\frac{k-\alpha\gamma}{k})^k]\Delta\rho_{opt}$, where $\Delta\rho_{DINO} := \rho(G) - \rho(\mathcal{G} \setminus \mathcal{L}_1)$, $\Delta\rho_{greedy} := \rho(G) - \rho(\mathcal{G} \setminus \mathcal{L}_2)$. γ and α are two constants defined as*

$$\gamma := \min_{|\Omega|=k, t\in\{0,\ldots,k-1\}} \frac{\Sigma_{w\in\Omega\setminus\mathcal{L}_1^t}(\hat{\rho}_{\mathcal{G}\setminus\mathcal{L}_1^t} - \hat{\rho}_{\mathcal{G}\setminus(\mathcal{L}_1^t\cup\{w\})})}{\hat{\rho}_{\mathcal{G}\setminus\mathcal{L}_1^t} - \hat{\rho}_{\mathcal{G}\setminus(\Omega\cup\mathcal{L}_1^t)}}, \quad (7)$$

where $\mathcal{L}_1^t$ is the removed node set by DINO with budget t, while

$$\alpha := \max_{\Omega, S\subset\mathcal{G}, |S|\leq k, |\Omega|=k, i\in S\setminus\Omega} (1 - \frac{\hat{\rho}_{G\setminus(S\setminus\{i\}\cup\Omega)} - \hat{\rho}_{(G\setminus(S\cup\Omega))}}{\hat{\rho}_{(G\setminus S)\cup\{i\}} - \hat{\rho}_{G\setminus S}}). \quad (8)$$

Proof. By Theorem 3.2 and the condition that all subgraphs of $\mathcal{G}$ up to removing the k nodes satisfy the three conditions given by Theorem 3.2, the spectral radius approximation given by $\hat{\rho}_{\mathcal{G}} = \frac{\mathbf{d}^{in}\mathbf{d}^{out}}{vol(\mathcal{G})}$ is precise enough (due to asymptotic convergence of the approximation function to the graph spectral radius) to guarantee that $\arg\max_{|\mathcal{L}|=k} \Delta\rho_{opt} = \arg\max_{|\mathcal{L}|=k} \Delta\hat{\rho}_{opt}$ and

$$\arg\max_{|\mathcal{L}|=k} \Delta\rho_{DINO} = \arg\max_{|\mathcal{L}|=k} \Delta\hat{\rho}_{greedy}. \quad (9)$$

This equation is establishes since $\lim_{N\to\infty} \rho_{DINO} = \rho_{DINO \text{ w/ } \rho}$, where "DINO w/ ρ" indicates the DINO algorithm with the node score function substituted with the spectral radius decrease caused by the node. Besides, since the greedy algorithm removes the node that causes the largest spectral radius decrease, and Theorem 3.1 claims that the node that causes the entire graph to have the largest spectral radius decrease is the node in the KSCC that causes the KSCC to have the largest spectral radius decrease, therefore we have $\rho_{DINO \text{ w/ } \rho} = \rho_{greedy}$.

Now, it suffices to estimate the approximation performance bound between the DINO and the global optimal algorithm. According to the Theorem 1 in [6], by assigning $F(\mathcal{L}) = \Delta\hat{\rho} := \hat{\rho}_{\mathcal{G}} - \hat{\rho}_{\mathcal{G}\setminus\mathcal{L}}$

and our defined γ, we have

$$\Delta\rho_{DINO} \geq \frac{1}{\alpha}\left[1 - (\frac{k-\alpha\gamma}{k})^k\right]\Delta\rho_{opt}, \tag{10}$$

The performance bound depends on the specific graph structures; we may employ $\hat{\rho}_{\mathcal{G}} = \frac{\mathbf{d}^{in}\mathbf{d}^{out}}{S}$ to γ and α and arrive at a theoretical approximation bound for the algorithm. We leave the development of more fine-grained bounds for γ and α for future work. □

In addition to effectiveness, we also present the efficiency of our DINO algorithm by proving its linear-time complexity. We omit the proof for brevity.

Theorem 4.2. ***(Efficiency)*** *Under mild conditions, the time complexity of Algorithm 1 is $O(k|V| + |E|)$.*

5 Experiments

In this section, we first introduce experimental settings, then we discuss the evaluation results of the proposed DINO. In particular, we aim to answer the following research questions: **RQ1:** How well can DINO decrease the network spectral radius and thus contain the epidemics compared with other baselines? **RQ2:** How efficient and scalable is DINO compared with other baselines? **RQ3:** How effective are each of the two components of DINO (*immunizing nodes within the KSCC* and *immunizing nodes using our node score function*) in network spectral radius decrease?

5.1 Experimental Settings

Datasets. We adopt six real-world directed networks of different sizes from various domains [26]. These networks include p2p-Gnutella (a file-sharing network), Wiki-Vote (a Wikipedia user voting network), Wiki-Talk (a Wikipedia user communication network), email-EuAll (an e-mail communication network), soc-Epin--ions1 (a user trust relationship network). Additionally, we perform experiment on the real-world epidemic dataset, called HIV, which is collected from the HIV transmission network of the U.S. (1988-2001) from the National Addiction & HIV Data Archive Program [33].

Baselines. In our work, we compare DINO with various epidemic containment methods as baselines. Specifically, the baselines include heuristic methods such as Random, Degree, Closeness Centrality [5], Katz Centrality [22], PageRank [36], and HITS [24], as well as state-of-the-art network epidemic containment methods Acquaintance [14], CBFinder [43] and CONTAIN [10]. Among those, CONTAIN [10] is a matrix perturbation-based epidemic containment method to validate the matrix perturbation methods are unsuitable for directed networks.

Evaluation Metrics. Based on Theorem 3.4, the effectiveness of a method in containing an epidemic is closely related to how well it can reduce the spectral radius of a network. Specifically, a larger spectral radius reduction indicates a more successful elimination of the epidemic. Therefore, we evaluate the performance of epidemic containment by analyzing the decrease in the spectral radius.

5.2 Effectiveness of DINO

We aim to answer **RQ1**. First, we compare our DINO algorithm with other baselines on six real-world unweighted directed networks. We perform experiments on each with the same node immunization budgets $\{100, 300, 500\}$. From Table 2, we can observe that DINO consistently outperforms all other baselines.

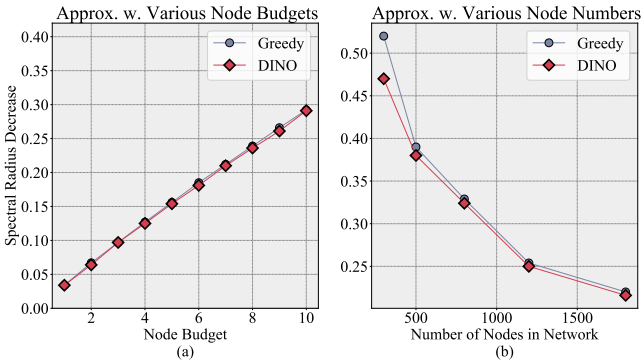


Figure 2: Performance approximation of DINO compared to Greedy algorithm in E-R networks.

In Fig. 2, encouraged by Theorem 4.1, we explore how closely DINO approximates the performance of the *Greedy algorithm*. The algorithm sequentially immunizes (i.e., removes) the node causing the largest spectral radius reduction. However, the Greedy algorithm is excessively time-consuming because it needs to compute the spectral radius after removing each candidate node. As a result, we limit experiments to networks of smaller sizes, specifically directed E-R networks, a special case of directed Chung-Lu networks where all nodes have the same expected node degrees [21]. In these networks, we can adjust the network size without affecting the properties of the epidemic spreading networks (as discussed in Section 2). In Fig. 2(a), we perform experiments on a directed E-R network of 1000 nodes with node immunization budgets from one to ten. We observe that on the same directed network, our DINO can approximate the performance of the Greedy algorithm under different budgets. In Fig. 2(b), we perform experiments on a directed E-R network with various node numbers $\{300, 500, 800, 1200, 1800\}$ and a fixed node immunization budget of 20. Increasing node numbers, the performance gap between DINO and the Greedy algorithm is reduced, which is consistent with Theorem 4.1 that DINO asymptotically approximates the performance of the Greedy algorithm, a ground truth that costs an extremely long time.

Finally, we perform an epidemic simulation on the Wiki-Vote dataset with three representative epidemic spread models (*SIS*, *SIR*, and *SEIR*). From the results in Fig. 3, the spread of the epidemic immunized by our DINO algorithm is contained more quickly than that of the baselines in all three epidemic models, especially *SIS*. Specifically, Fig. 3(a) shows the infected people of different time steps under *SIS* epidemic model. We can observe that DINO eliminates the epidemic at time step 14, while the second effective method takes until time step 67.

5.3 Efficiency and Scalability

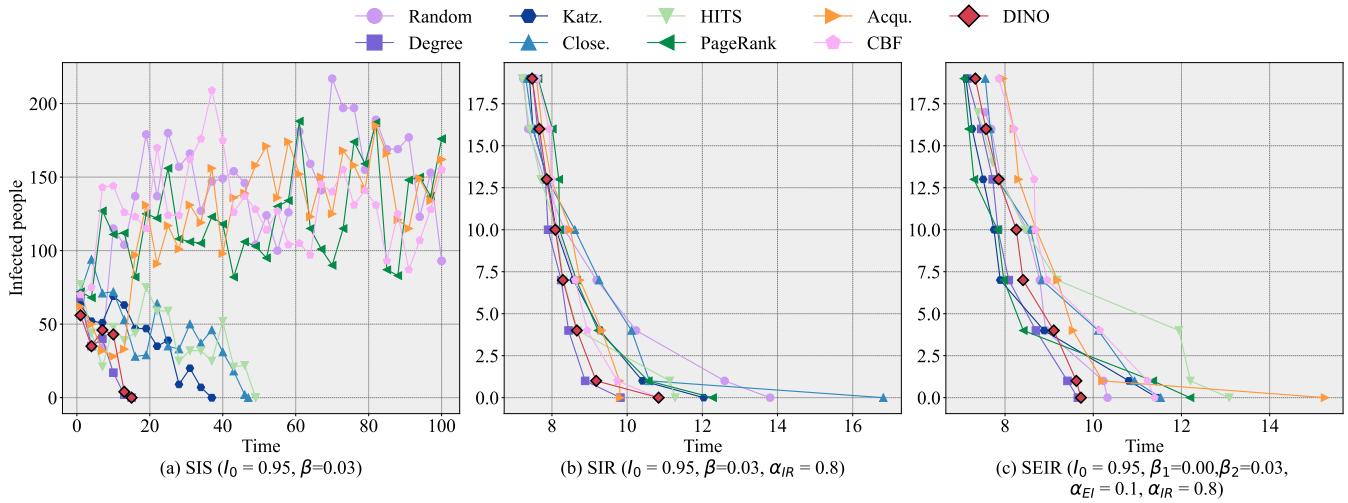
We aim to answer **RQ2**. We evaluate the efficiency and scalability of our proposed DINO algorithm and present the result in Fig. 4. Fig. 4(a) compares the quality vs. running time trade-off of DINO with baselines. We conduct DINO and other baselins on p2p-Gnutella with an immunization node budget 20. DINO is orders of magnitude

Table 1: The basic information about the real-world datasets adopted for experimental evaluation. KSCC denotes the number of nodes in the key strongly connected component.

Dataset	p2p-Gnutella	Wiki-Vote	soc-epinions1	email-EuAll	web-BerkStan	Bitcoin-Alpha	Bitcoin-OTC
Network Type	Pear-to-peer	Social	Social	Communication	Web	Bitcoin Trade	Bitcoin Trade
$\|V\|$	6,301	7,115	75,879	265,214	685,231	3,783	5,881
$\|E\|$	20,777	103,689	508,837	420,045	7,600,595	24,186	35,592
KSCC	2,068	1,300	32,223	34,203	334,857	3,235	4,709

Table 2: Experimental results of spectral radius decrease caused by immunization node set on six real-world directed networks under different budgets. Here ρ means the spectral radius of the directed network, and k is the node budget. "OOT" stands for "Out of Time"(experiment exceeds 48 hours).

Dataset	p2p-Gnutella			Wiki-Vote			email-EuAll			soc-Epinions1			Wiki-Talk			HIV		
ρ	5.12			45.14			69.53			106.53			228.94			7.32		
k	100	300	500	100	300	500	100	300	500	100	300	500	100	300	500	100	300	500
Random	0.03	0.07	0.12	1.02	2.00	3.16	0.0006	0.0028	0.0068	0.06	0.12	0.31	0.001	0.0061	0.0061	0.05	0.05	0.05
Degree	1.67	2.26	2.62	23.10	35.23	40.00	19.31	46.96	57.55	33.12	59.54	60.05	26.22	69.06	105.73	1.05	2.48	3.25
KatzCen.	1.53	1.90	2.10	19.22	26.65	30.39	25.51	48.49	55.29	34.57	58.59	59.67	62.61	118.58	147.64	3.36	4.02	4.02
CloseCen.	1.50	1.64	1.82	17.09	23.29	24.40	34.91	48.68	51.47	27.10	50.30	59.43	OOT	OOT	OOT	1.04	1.04	1.04
HITS	0.64	1.32	1.69	19.99	31.72	37.31	0.51	0.51	0.66	30.72	51.29	60.87	58.78	115.33	142.80	2.01	2.01	3.19
PageRank	1.37	1.44	1.44	14.67	21.62	26.70	25.57	48.49	54.80	28.42	54.55	59.37	43.96	98.14	127.89	0.00	0.00	0.00
Acqua.	0.62	0.95	1.43	7.64	12.03	17.52	5.81	23.17	30.38	3.58	10.80	16.46	7.34	21.64	35.86	0.09	0.14	0.15
CBFinder	1.54	2.09	2.43	6.15	20.31	31.17	12.09	30.30	47.72	8.71	16.20	24.63	3.76	15.59	17.92	1.12	2.66	3.78
CONTAIN	0.56	1.84	2.18	23.13	36.05	38.90	OOT	OOT	OOT	37.32	55.77	60.63	OOT	OOT	OOT	1.04	OOT	OOT
DINO	**2.83**	**5.12**	**5.12**	**26.88**	**38.81**	**43.14**	**34.17**	**55.24**	**62.02**	**40.59**	**60.90**	**70.82**	**65.28**	**122.05**	**150.94**	**4.59**	**5.70**	**6.32**



(a) SIS (I_0 = 0.95, β=0.03) (b) SIR (I_0 = 0.95, β=0.03, α_{IR} = 0.8) (c) SEIR (I_0 = 0.95, β_1=0.00,β_2=0.03, α_{EI} = 0.1, α_{IR} = 0.8)

Figure 3: Epidemic simulation results in the Wiki-Vote network using *SIS, SIR, SEIR* epmidemic models with a node immunization budget of 20. For better visibility, the x-axis does not start at zero.

faster than the Greedy algorithm while achieving similar performance. Although the running time of other baseline methods is similar to DINO, their performance (y-axis) are incomparable with DINO. Fig. 4(b) shows our DINO's scalability with the increase in network size by conducting our DINO algorithm on directed E-R networks [35] with different sizes. The proposed DINO algorithm scales linearly with respect to network size (that is, the sum of the number of nodes and edges), consistent with Theorem 4.2.

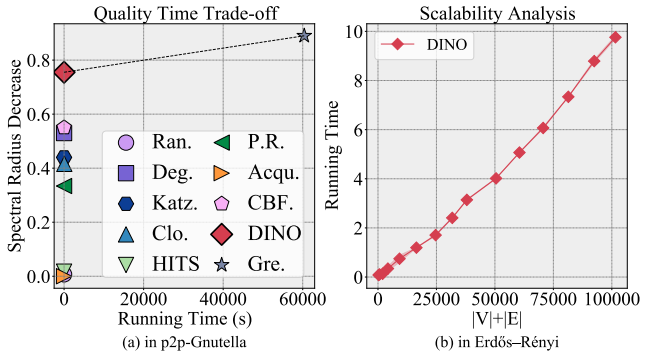


Figure 4: Efficiency and scalability evaluation of the DINO Algorithm. "Ran.", "Deg.", "Katz.", "Clo.", "P.R.", "Acqu.", "CBF." represent Random, Degree, Katz Centrality, Closeness Centrality, PageRank, Acquaintance, and CBFinder algorithm, respectively. Node immunization budget is 20.

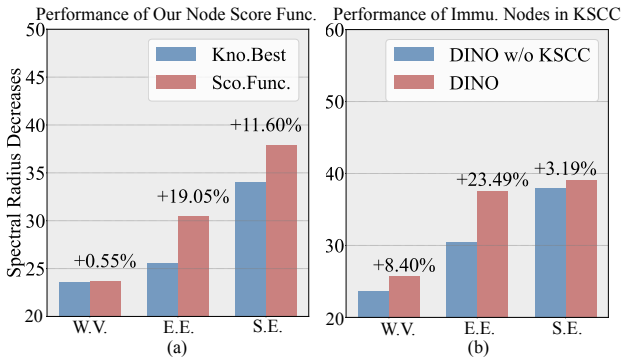


Figure 5: Detailed study of DINO. "Sco.Func." immunizes nodes with the highest proposed node score. "Kno.Best" is the maximum spectral radius reduction among Katz Centrality, Closeness Centrality, Degree, and HITS. "W.V.", "E.E.", and "S.E." represents Wiki-Vote, email-EuAll, and soc-Epinions1.

5.4 Detailed Study

We answer **RQ3** by evaluating DINO's two components: *Immunizing node within KSCC* and *Immunizing Node with Designed Node Score Function*. Fig. 5 shows the network spectral radius reduction in three real-world networks (Wiki-Vote, email-EuAll, and soc-Epinions1) with a node immunization budget 100.

Fig. 5(a) shows the performance of our designed node score function compared to baseline functions (Degree, Katz Centrality, Closeness Centrality, and HITS score). Removing nodes with ours results in a greater spectral radius reduction than the baselines, especially on large networks like Wiki-Talk, proving its effectiveness as an approximation for spectral radius decrease. Fig. 5(b) compares spectral radius reduction of DINO with and without immunizing nodes within the KSCC. Immunizing nodes within the KSCC significantly improves the algorithm's performance (up to 23.49%). Hence, removing nodes within the KSCC enhance epidemic containment.

6 Related Work

6.1 Epidemiology on Directed Networks

Recent studies in network epidemiology [16] involving directed networks have been extensions of *Individual-based Mean Field Theory* (IBMF) [9, 47] or *Degree-based Mean Field Theory* (DBMF) [4, 18]. IBMF and DBMF are fundamental theories for undirected networks. IBMF assumes the dynamic state of every node is statistically independent of its neighbors. [37] adapted IBMF to directed networks for the *SIR* model, finding the epidemic threshold is negatively correlated with the proportion of bidirectional edges [29]. DBMF assumes nodes of the same degree are statistically equivalent. Based on DBMF, the epidemic threshold is closely related to the spectral radius of the network's extended connectivity matrix for *SIS* and *SIR* model [7, 45]. Most prior works focus on a specific epidemic model, while we try to understand epidemic spread with a more general model, $S^* I^2 V^*$, capturing most epidemics in directed networks.

6.2 Node Immunization

Node immunization algorithms can be classified into *structural-based* and *spectral-based* methods. Structural-based methods analyze the relationship between epidemic spread and network features, such as connectivity and centrality. Selecting nodes with the highest betweenness centrality [20] is a popular approach. Other strategies include connected cluster identification [44] and node influence estimation [38]. Spectral-based methods aim to immunize nodes that significantly decrease the network's spectral radius [47]. When the spectral radius is smaller than the epidemic threshold, the epidemic will be eliminated. The most common way to reduce the spectral radius is to apply matrix perturbation tools to the adjacency matrix [10, 11, 46]. Matrix perturbation-based methods are effective but cannot be applied to directed networks. This is because their asymmetric adjacency matrices' eigenvectors do not form an orthogonal basis, and their eigenvalues are not Lipschitz continuous under matrix perturbations.

7 Conclusion

Matrix perturbation is inapplicable for directed network epidemic containment problem. In this work, We take initial steps toward epidemic containment problems on directed networks without matrix perturbation theory. Specifically, we first derive the epidemic threshold for a general epidemic spread model on directed networks, then we present a theorem to estimate the spectral radius of directed networks. Based on our theoretical results, we propose DINO algorithm which immunizes a limited number of nodes to contain an epidemic. Extensive experiments demonstrate its effectiveness and efficiency compared with state-of-the-art baselines on various real-world epidemic spread networks.

8 Acknowledgement

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9 Ethical Considerations

The implementation of our epidemic containment strategies on directed networks necessitates access to detailed network data, often including sensitive personal information. This raises significant privacy and fairness concerns, particularly in social network analysis and healthcare applications. Any users employing our algorithm should follow relevant privacy regulations when performing experiments on real-world datasets. Despite under the regulated scale, we still encourage users to collect only the minimum necessary data for the task. Additionally, due to demographic differences, individual information may be unfairly collected, resulting in a biased epidemic network. Thorough bias audits should be conducted, analyzing selected nodes for patterns of unfair treatment across protected attributes.

Our DINO method could potentially be misused for malicious purposes, such as targeting individuals for disinformation campaigns or disrupting social networks. To address this issue, researchers should implement tools to safeguard our algorithm against misuse, including technical measures like access controls and usage monitoring for software implementations of the DINO algorithm. Licensing agreements should explicitly prohibit malicious applications. Increased awareness and education within the research community about the dual-use nature of network analysis techniques is necessary, with ethics training as a mandatory component of research in this field. Collaboration with policymakers and ethicists is crucial to develop appropriate governance frameworks, potentially including ethics review boards for network analysis research or industry-wide best practices. Researchers should also consider implementing "ethical exit" strategies which design predetermined plans for suspending or terminating research if evidence of misuse or unintended harmful consequences emerges.